Article

KAN-Payne: A Kolmogorov–Arnold Network Spectral Emulator for Stellar Label Inference

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Abstract

Stellar labels in large spectroscopic surveys are commonly obtained by matching observed spectra to synthetic grids or by training empirical spectral models on reference stars. Both strategies depend on a fast representation of the mapping between stellar labels and flux. We present KAN-Payne, a Payne-style spectral emulator that replaces the fully connected multilayer perceptron in The Payne with a Kolmogorov–Arnold Network. Stellar spectra are smooth functions of atmospheric labels over much of the training domain, while individual wavelength pixels can have sharply varying responses near temperature- and abundance-sensitive features. KAN layers, with learnable univariate functions on network edges, provide a direct way to model such responses. We evaluate KAN-Payne on the APOGEE-like Kurucz synthetic spectra released with The Payne and use Payne-MLP and a TransformerPayne-style model as reference baselines. The benchmark fixes the wavelength grid, pixel mask, label list, training–validation split, and label scaling, so the comparison focuses on emulator architecture. On the 10000-epoch synthetic-grid benchmark, KAN-Payne reaches a validation MAE of 20.89 in units of 10^4 times the flux residual, compared with 57.71 for Payne-MLP and 35.18 for the TransformerPayne-style baseline. The public pretrained Payne MLP reaches 17.44 on the same file and is retained as an external reference because its training protocol differs. We then use the trained emulators as differentiable forward models for APOGEE DR17 spectra. On a common 9117-star quality-controlled sample, KAN-Payne gives a robust scatter of 113 K in $T_{\rm eff}$ and 0.053 dex in [Fe/H] relative to ASPCAP, smaller than the corresponding Payne-MLP and TransformerPayne-style values for these two labels. The main observed-spectrum systematic is a positive $\log g$ offset. The paper is organized as a methods study and emphasizes the accuracy and deployment of KAN-Payne for survey-scale stellar parameter inference.

Keywords: stellar spectroscopy; APOGEE; stellar parameters; Kolmogorov–Arnold networks; spectral emulators; machine learning

Received:
Revised:
Accepted:
Published:

Citation: Zhang, S. & Wang, R. KAN-Payne: A Kolmogorov–Arnold Network Spectral Emulator for Stellar Label Inference. *Universe* **2026**, *1*, 0.

1. Introduction

Stellar spectra are one of the main observational links between Galactic surveys and stellar astrophysics. Effective temperature, surface gravity, metallicity, and detailed abundance ratios enter distance estimation, age inference, chemical-tagging studies, selection-function modeling, and the construction of chemo-dynamical maps of the Milky Way. The scientific value of large spectroscopic surveys therefore depends not only on the number of spectra, but also on the reliability and homogeneity of the stellar labels derived from them.

This requirement has shaped the analysis systems of modern surveys such as APOGEE, LAMOST, GALAH, and RAVE [1–5].

Most production pipelines still start from a physically interpretable idea: compare the observed spectrum with synthetic spectra and find the atmospheric labels that minimize a spectral residual. APOGEE's ASPCAP performs pseudo-continuum normalization, fits global stellar parameters with synthetic model grids, and then derives individual abundances in selected wavelength windows [6]. FERRE supplies the grid interpolation and optimization machinery used by ASPCAP and related analyses [7]. LAMOST uses automated low-resolution pipelines such as LASP and LSP3 to deliver radial velocities and atmospheric parameters for millions of stars [3,8,9]. RAVE DR4 combined MATISSE and DEGAS for medium-resolution Ca-triplet spectra [5]. GALAH DR3 used Spectroscopy Made Easy with MARCS model atmospheres, while later GALAH analyses increasingly combine synthetic spectra with neural-network acceleration [4]. These pipelines are successful because they keep contact with stellar-atmosphere physics, but their cost grows quickly with the number of labels, wavelength pixels, and survey spectra.

A second family of methods learns the spectrum–label relation from reference stars. The Cannon models normalized flux at each wavelength as a flexible function of stellar labels, enabling label transfer between surveys [10,11]. StarNet and astroNN showed that neural networks can infer APOGEE-like parameters and abundances rapidly from observed spectra [12,13]. SLAM used support-vector regression for LAMOST spectra, and DD-Payne combined empirical training labels with Payne-style constraints to estimate abundances for millions of LAMOST stars [14,15]. These approaches are practical and fast, but their calibration is tied to the reference set: outside the training domain, extrapolation and inherited label systematics become central concerns.

The Payne occupies a useful middle ground. It trains a neural emulator on synthetic spectra, then uses the differentiable emulator as a forward model for observed spectra [16]. This keeps the model anchored to a physical spectral grid while avoiding repeated high-dimensional interpolation during fitting. TransformerPayne recently introduced a wavelength-conditioned transformer emulator and showed that attention-based models can improve emulation accuracy and data efficiency in synthetic-grid tests [17]. The remaining question is whether there is a simpler vector-output architecture that keeps the deployment advantages of The Payne while improving over the standard multilayer perceptron.

This paper introduces KAN-Payne, a Payne-style spectral emulator based on Kolmogorov–Arnold Networks. KANs replace fixed node activations with learnable univariate functions on network edges [18]. For stellar spectra this is a natural parameterization: the flux response to atmospheric labels is smooth over much of the grid, but individual pixels can change sharply near temperature-sensitive molecular bands or abundance-sensitive atomic lines. KAN-Payne therefore targets a specific niche. It is not proposed as a universal replacement for survey pipelines or for transformer models. Instead, it is designed as a full-spectrum, differentiable emulator that can be dropped into a Payne-like fitting loop.

This work first formulates KAN-Payne as a vector-output KAN spectral emulator for Payne-style stellar label inference. It then builds a controlled benchmark on the public Payne APOGEE/Kurucz synthetic training set, with the wavelength grid, label scaling, validation split, and pixel mask held fixed across all models. Under this matched setup, KAN-Payne is compared with Payne-MLP and TransformerPayne-style baseline through training curves, validation residual distributions, and wavelength-resolved residuals. The same emulators are finally used in an APOGEE DR17 parameter-inference experiment, where fitted labels from observed spectra are compared with the official ASPCAP catalog.

2. Data

2.1. The Payne Synthetic Spectral Grid

The emulator benchmark uses the Kurucz APOGEE-like synthetic spectra released with The Payne GitHub repository [16]. The file contains 1000 continuum-normalized spectra on the Payne APOGEE wavelength grid. After loading the public wavelength and mask files, the working grid has 7214 pixels from 15168.13 to 16936.75 Å. The target vector for each spectrum contains 25 labels: $T_{\rm eff}$, $\log g$, microturbulence, 19 elemental abundance labels, ^{12}C/^{13}C, and macroturbulence. We keep the repository split: 800 spectra for training and 200 spectra for validation. Pixels marked by the Payne APOGEE mask are excluded from the primary validation metric, but they are kept in the arrays so that all models return spectra on the same grid.

The labels are scaled using training-set bounds,

$$\tilde{\ell}_i = \frac{\ell_i - \ell_{i,\rm min}}{\ell_{i,\rm max} - \ell_{i,\rm min}} - 0.5, \tag{1}$$

which maps the training range to approximately $[-0.5, 0.5]$. The spectra are used in their continuum-normalized flux units. For the vector-output models, the loss is computed across the full 7214-pixel vector. For the TransformerPayne-style baseline, wavelengths are additionally scaled to $[-1, 1]$ because the model predicts flux as a function of labels and wavelength.

2.2. APOGEE DR17 Spectra and ASPCAP Labels

The observed-spectra experiment uses APOGEE DR17 `aspcapStar` products and the official `allStar-dr17-synspec_rev1` catalog. APOGEE observes near-infrared H-band spectra at a resolving power of approximately 22500 over 1.51–1.70 μm [1,19]. DR17 is the final SDSS-IV release and includes the completed APOGEE-2 data set [2]. ASPCAP provides the comparison labels used here: calibrated $T_{\rm eff}$, $\log g$, [M/H], [Fe/H], [α/M], and selected abundance ratios [6,20].

We select a conservative red-giant clean sample from the DR17 catalog. The selection requires finite ASPCAP labels, no severe ASPCAP or stellar warning flags, at least two visits, APOGEE spectra from either the APO or LCO telescope, $3500 < T_{\rm eff} < 5500$ K, $0 < \log g < 3.8$, $-2.0 <$ [Fe/H] < 0.5, and `SNREV`> 100. The current analysis sample contains 10000 stars. Its median ASPCAP labels are $T_{\rm eff} = 4658$ K, $\log g = 2.39$, and [Fe/H] $= -0.23$. The median per-star S/N estimated from the normalized flux and uncertainty arrays is 137.

2.3. Observed-Spectrum Preprocessing

We work with `aspcapStar` spectra rather than raw visit spectra. These files are already combined, shifted to the stellar rest frame, and supplied with the APOGEE pseudo-continuum normalization. However, direct use of the `aspcapStar` flux leaves chip-scale continuum residuals that are not well matched to the Payne synthetic spectra. We therefore follow the normalization strategy used in The Payne fitting code for APOGEE spectra: after interpolation onto the Payne wavelength grid, a fourth-order Chebyshev continuum is fitted separately on the blue, green, and red APOGEE detector segments using a fixed set of continuum pixels. In the absence of the original binary continuum-pixel file, we rebuild an equivalent mask from the public Payne synthetic grid by selecting, within each detector segment, pixels with median synthetic flux closest to unity and low object-to-object variance. The same continuum-pixel mask is then used for every APOGEE spectrum.

The preprocessing begins by downloading the selected `aspcapStar` files from the SDSS DR17 SAS tree. For each FITS file we read the normalized flux, uncertainty, best-fit ASPCAP model, and wavelength solution from the image HDUs. A 0.005 floor in normalized-flux units is added to the flux uncertainty before fitting so that individual pixels with very small formal errors do not dominate the likelihood. Flux and uncertainty are then resampled from the 8575-pixel APOGEE `aspcapStar` grid to the 7214-pixel Payne grid by one-dimensional interpolation. After resampling, the three-chip Chebyshev continuum described above is fitted and divided out using identical continuum pixels for the full sample. The final fitting mask combines the APOGEE bad-pixel mask, non-finite values, non-positive uncertainties, and the Payne pixel mask. The median number of usable Payne-grid pixels per star after masking is 6339, about 87.9% of the 7214-pixel grid. The continuum-pixel mask contains 576 pixels, distributed as 233, 192, and 151 pixels across the three APOGEE detector segments. This step changes only the smooth continuum scale; the line-level mask and ASPCAP labels are unchanged.

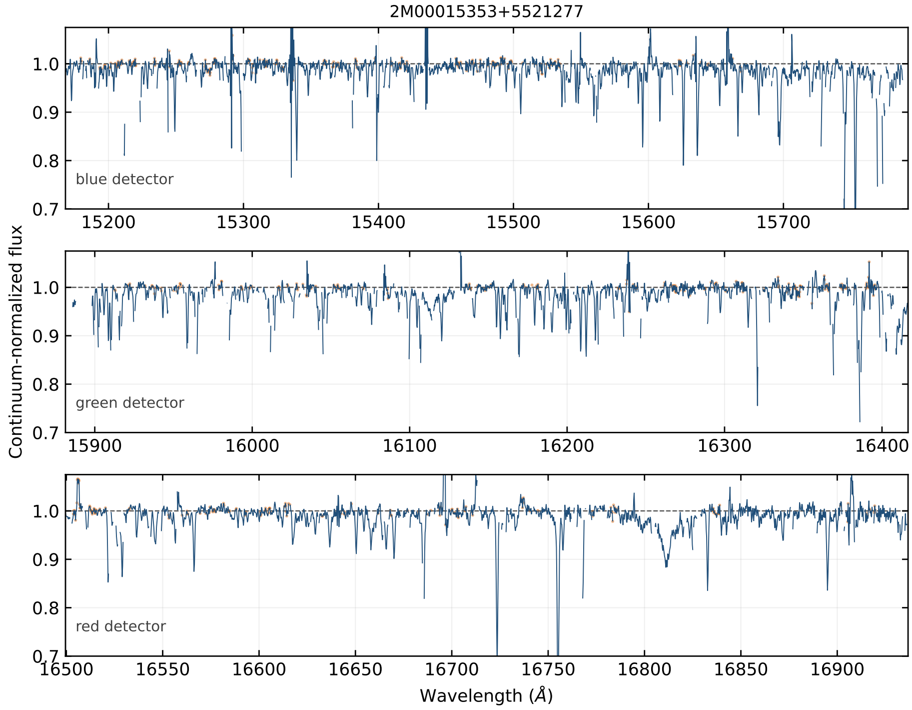

Figure 1. Example APOGEE DR17 `aspcapStar` spectrum after interpolation onto the Payne wavelength grid and Payne-style three-chip Chebyshev continuum renormalization. The three panels show the blue, green, and red APOGEE detector segments separately so that detector gaps are not connected by the plotted line. Orange markers indicate the fixed continuum pixels used for the Chebyshev fit.

3. Methods

3.1. Payne-MLP Baseline

The Payne-MLP baseline maps the 25 scaled stellar labels directly to the full 7214-pixel spectrum through two hidden layers. We reproduce the original GitHub-released pretrained Payne network with a NumPy forward pass and train a new MLP under the same unified split for a controlled comparison.

For the controlled run the network has two hidden layers with 300 neurons each. The activation function is a LeakyReLU with negative slope 0.01. The model is trained with a mean absolute flux loss,

$$\mathcal{L}_{\mathrm{MLP}} = \frac{1}{N_{\mathrm{batch}} N_{\mathrm{pix}}} \sum_{n,p} |f_{n,p}^{\mathrm{pred}} - f_{n,p}^{\mathrm{true}}|, \tag{2}$$

using raw continuum-normalized fluxes rather than per-pixel normalized targets. Validation is reported in units of 10^4 times the flux residual. The formal 10000-epoch run uses AdamW with learning rate 10^{-4}, zero weight decay, no target-spectrum normalization, and the best checkpoint selected by validation MAE on unmasked Payne pixels. The Payne-MLP uses batch size 128. KAN-Payne and the TransformerPayne-style baseline use the same optimizer settings with batch sizes 64 and 32, respectively, chosen to keep the A100 training jobs stable.

3.2. KAN-Payne: Proposed Emulator

KAN-Payne replaces the MLP layers with a vector-output KAN emulator. The model uses the same input labels, target spectra, loss function, and validation split as Payne-MLP. This isolates the effect of the KAN architecture from data and preprocessing choices.

The architectural difference from a standard MLP is easiest to see at the level of one layer. In an MLP, each hidden unit first forms a linear mixture of the previous activations and then applies a fixed scalar nonlinearity,

$$\mathbf{h}^{(l+1)} = \sigma\Big(\mathbf{W}^{(l)}\mathbf{h}^{(l)} + \mathbf{b}^{(l)}\Big), \tag{3}$$

where the trainable part is the affine map and the activation function σ is fixed in advance. This design is expressive and computationally efficient, but all nonlinear structure must be assembled by combining many fixed activations. A KAN layer instead assigns a learnable univariate function to each edge between input and output coordinates,

$$h_j^{(l+1)} = \sum_i \phi_{ji}^{(l)}\Big(h_i^{(l)}\Big), \tag{4}$$

where the functions $\phi_{ji}^{(l)}$ are trainable. In practice these edge functions are represented by a smooth basis expansion together with a simple base activation. The model therefore learns not only how strongly one label coordinate contributes to the next representation, but also the shape of that one-dimensional response.

This distinction is important for stellar spectra. The flux at a given pixel is often a smooth but non-linear function of $T_{\rm eff}$, $\log g$, and abundance labels. The response can be nearly monotonic in continuum regions, change rapidly near ionization or molecular-equilibrium transitions, and become locally saturated in strong absorption lines. An MLP can model these effects, but it does so through repeated affine mixing followed by fixed nonlinearities. KAN-Payne gives each edge a flexible response curve, which is a more direct parameterization for the smooth one-dimensional label dependencies that appear throughout a synthetic spectral grid. The summation over learned edge functions also keeps the model differentiable with respect to the input labels, which is required by the downstream Payne-style fitting step.

The adopted architecture is

$$25 \to 64 \to 256 \to 7214, \tag{5}$$

with GELU base activations in the KAN layers. The implementation uses the efficient-kan edge-function parameterization, in which each edge function combines a cubic spline component and a base activation. We use `grid_size=5`, `spline_order=3`, the default efficient-kan noise and spline scaling, and GELU as the base activation. The input dimension is the 25 scaled Payne labels, and the output dimension is the full 7214-pixel APOGEE-like spectrum. We use a vector-output design rather than training one independent model per wavelength pixel. This couples the pixel predictions through shared hidden representations and allows a single forward pass to produce the entire spectrum. The architecture is not meant to be the final architecture search for KANs; rather, it is a practical configuration that can be trained repeatedly while preserving enough capacity to model thousands of APOGEE pixels.

KAN-Payne keeps the main advantages of The Payne formulation. It predicts the whole spectrum in one forward pass, it is differentiable with respect to the stellar labels, and it can therefore be used directly inside gradient-based spectral fitting. Its distinction from Payne-MLP lies in the parameterization of the label–flux function: instead of relying only on affine transformations and fixed pointwise activations, each KAN layer learns edge functions whose shapes can adapt to the training grid. Compared with wavelength-conditioned TransformerPayne-style models, KAN-Payne does not explicitly condition on wavelength and therefore lacks the transformer's long-range attention mechanism, but it is simpler to deploy in a full-spectrum fitting loop because it retains vector-output behavior. In this work, KAN-Payne is therefore treated as a vector-output alternative to the original Payne-MLP: more adaptive than a fixed-activation MLP, but still close to The Payne in how it is called during spectral fitting.

3.3. TransformerPayne-Style Baseline

Our TransformerPayne-style baseline is implemented as a wavelength-conditioned scalar-flux model. It encodes the label vector into learned label tokens and encodes the requested wavelength with sinusoidal features. Cross-attention blocks condition wavelength queries on the label tokens, and a small regression head predicts the flux at the requested wavelength. During training we sample wavelength pixels for each spectrum; during validation we scan the full 7214-pixel grid in chunks. This implementation follows the main architectural idea of TransformerPayne but is trained inside the same data loader, optimizer schedule, and validation framework as the MLP and KAN models. It should therefore be read as a local TransformerPayne-style comparison rather than a claim about every possible tuning of the published TransformerPayne code.

For the main experiment the label vector is mapped to 16 tokens of width 128. The wavelength coordinate is scaled to $[-1, 1]$ and expanded with sinusoidal features. Four cross-attention blocks with four attention heads update the wavelength query against the label tokens. At each training step, 256 wavelength pixels are sampled per spectrum. This sampling scheme makes the scalar-output model compatible with the same validation protocol while still evaluating the full spectrum during validation.

3.4. APOGEE Parameter Inference

For each observed APOGEE spectrum we optimize the 25 scaled Payne labels by minimizing an uncertainty-weighted residual objective,

$$\chi^2_{\mathrm{red-like}}(\boldsymbol{\ell}) = \frac{1}{N_{\mathrm{pix}}} \sum_{p \in \mathcal{P}} \left[\frac{f_p^{\mathrm{obs}} - f_p^{\mathrm{emu}}(\boldsymbol{\ell})}{\sigma_p} \right]^2, \tag{6}$$

where $\mathcal{P}$ excludes masked Payne pixels and bad observed pixels. The normalization by N_{pix} makes this a reduced-chi-square-like objective rather than a formally calibrated reduced χ^2, because the continuum-normalized pixels have correlated errors and an imposed uncertainty floor. ASPCAP labels initialize T_{eff}, $\log g$, and [Fe/H] when available; other labels start from the training-grid midpoint unless an ASPCAP abundance proxy is available. The fitted labels are then compared against ASPCAP calibrated parameters.

4. Results

4.1. Synthetic Emulator Benchmark

All emulator comparisons use the same 800/200 synthetic split. We report MAE and RMSE in units of flux residuals multiplied by 10^4, both over all pixels and over unmasked Payne pixels. The pretrained Payne-MLP released with The Payne gives a validation MAE of 17.44 and RMSE of 68.31 on unmasked pixels for this repository benchmark. This pretrained model is not a controlled baseline because its training details and stopping point differ from our run; it is kept as an external reference to check that the data loader and wavelength grid are consistent with the public Payne release.

The validation figures follow the convention used in survey-pipeline and spectral-emulator papers: a scalar summary table is not sufficient by itself. We therefore use four complementary views. The first is the training history, which shows whether a model has reached a stable validation floor or is still improving. The second is the residual histogram over all validation spectra and unmasked pixels, which makes non-Gaussian tails visible. The third is the wavelength-resolved residual curve, which shows whether an emulator fails globally or only near specific spectral features. The fourth is a star-by-star residual summary, used to identify whether a small subset of validation spectra dominates the metric. This style follows the logic of ASPCAP validation, The Payne emulator tests, and neural-network label-transfer studies: scalar precision, systematic residuals, and outlier structure are reported together [6,12,13,16].

Table 1. Synthetic validation performance. KAN-Payne is the proposed method; Payne-MLP and the TransformerPayne-style emulator define the local comparison baseline. The trainable-model rows use the best validation checkpoints from the 10000-epoch A100 run.

Model	Epoch	$\mathrm{MAE}_{\mathrm{good}}$	$\mathrm{RMSE}_{\mathrm{good}}$	Notes
Payne pretrained MLP	–	17.44	68.31	GitHub coefficients
Payne-MLP	9933	57.71	106.81	best 10k checkpoint
KAN-Payne	9964	20.89	57.12	best 10k checkpoint
TransformerPayne-style	9932	35.18	86.94	best 10k checkpoint

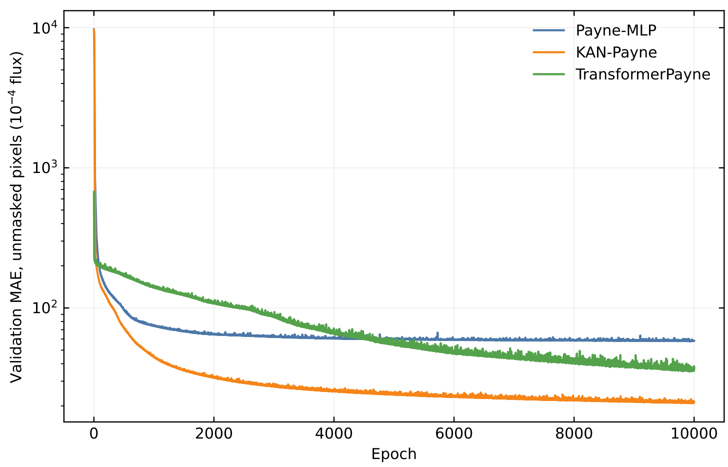

Figure 2. Validation MAE on unmasked Payne pixels as a function of training epoch. KAN-Payne reaches the lowest validation floor among the three controlled emulator runs.

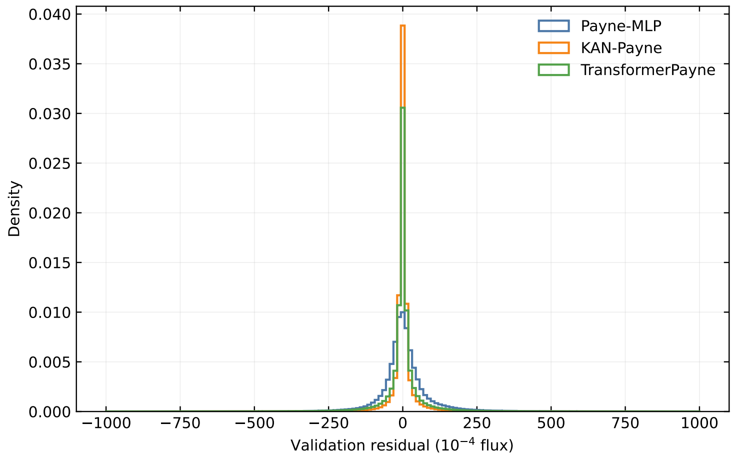

Figure 3. Distribution of validation residuals over all validation spectra and unmasked Payne pixels. KAN-Payne narrows the residual core relative to the controlled Payne-MLP and TransformerPayne-style runs.

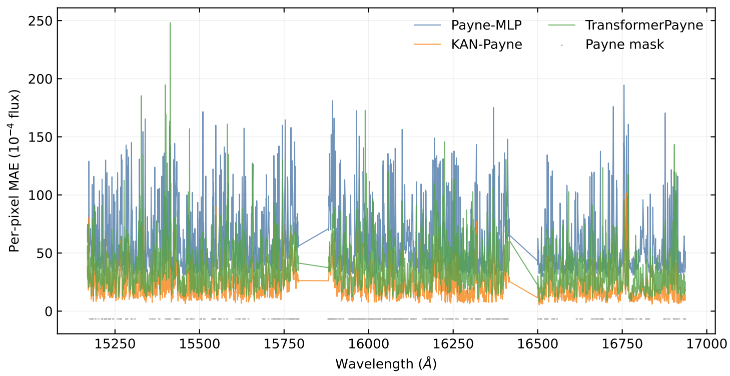

Figure 4. Per-pixel validation residuals for Payne-MLP, KAN-Payne, and the TransformerPayne-style baseline. The plot highlights wavelength regions where emulator errors are concentrated.

KAN-Payne reaches a validation MAE of 20.89 and RMSE of 57.12 on unmasked Payne pixels. Under the same split and flux target, Payne-MLP reaches 57.71 and 106.81, while the TransformerPayne-style baseline reaches 35.18 and 86.94. KAN-Payne therefore reduces the MAE by 63.8% relative to Payne-MLP and by 40.6% relative to the TransformerPayne-style baseline in this synthetic validation experiment. The wavelength-resolved plot shows that the improvement is not confined to a single small spectral interval; KAN-Payne lowers the residual level across much of the Payne grid, while the largest remaining errors still occur near strong and blended spectral features. The pretrained Payne MLP remains the best absolute reference on this public 1000-spectrum file under the original Payne training protocol, so the claim made here is restricted to the controlled local training protocol.

4.2. APOGEE DR17 Stellar Parameter Inference

The APOGEE DR17 experiment asks whether the synthetic-emulator ranking carries over to observed spectra. We fit the same 10000-star clean sample with Payne-MLP, KAN-Payne, and the TransformerPayne-style baseline. For each spectrum we optimize the 25 scaled Payne labels by minimizing the uncertainty-weighted residual objective in Equation (6). The fit uses 2048 good pixels sampled without replacement for optimization; the sampling uses a global random seed of 42, is repeated identically for the three emulators, and is held fixed for each star during its 600 Adam optimization steps. The final residual is then evaluated on all unmasked pixels. ASPCAP values initialize $T_{\rm eff}$, $\log g$, [Fe/H], and the available abundance proxies; labels without an ASPCAP proxy start at the center of the synthetic training range. Each model is fitted from its best 10000-epoch synthetic-validation checkpoint.

The reported APOGEE metrics are the median residual, the robust scatter 1.4826 MAD, and the 16th–84th percentile interval of fit–ASPCAP residuals for $T_{\rm eff}$, $\log g$, and [Fe/H]. We also track the median spectral fit MAE and $\chi^2_{\rm red-like}$ because a model can agree with ASPCAP labels while still fitting the normalized spectrum poorly. All three models successfully return labels for all 10000 spectra. For the atmospheric comparison we require at least 5000 valid pixels, reject the upper 1% tail in model-specific $\chi^2_{\rm red-like}$ and spectral MAE, and require the three atmospheric labels to remain inside the scaled training interval. These model-specific quality cuts keep 9244 Payne-MLP stars, 9526 KAN-Payne stars, and 9408 TransformerPayne-style stars. To avoid comparing different stellar samples, the reported residual table and the main APOGEE figures use the intersection of the three quality masks, containing 9117 stars. On this common sample the median spectral MAE values are 148.7, 147.4, and 166.3 in units of 10^4 times the flux residual for Payne-MLP, KAN-Payne, and the TransformerPayne-style baseline, respectively. The spectral MAE values for Payne-MLP and KAN-Payne are therefore close, while their atmospheric residual scatters differ more strongly. This is expected because the fit objective sums over thousands of continuum-normalized pixels, whereas the reported labels are most sensitive to particular temperature-, gravity-, and metallicity-sensitive features.

The same conclusion holds on the model-specific quality samples. The own-sample $T_{\rm eff}$/[Fe/H] scatters are 125.5 K/0.0888 dex for Payne-MLP, 113.3 K/0.0534 dex for KAN-Payne, and 137.4 K/0.0615 dex for the TransformerPayne-style baseline, close to the corresponding common-sample values in Table 2. Thus, the common-sample intersection does not artificially create the KAN-Payne advantage by selecting only stars that are easy for all three emulators.

Table 2. APOGEE DR17 atmospheric-label comparison against ASPCAP on the common 9117-star quality-controlled sample. Residuals are defined as fitted label minus ASPCAP label. The scatter column gives 1.4826 MAD.

Model	Label	N	Median	1.4826 MAD	P16–P84
Payne-MLP	$T_{\rm eff}$ (K)	9117	-62.9	124.8	-182.0–78.0
KAN-Payne	$T_{\rm eff}$ (K)	9117	-17.4	112.9	-120.9–108.2
TransformerPayne-style	$T_{\rm eff}$ (K)	9117	-47.1	136.4	-193.5–83.8
Payne-MLP	$\log g$ (dex)	9117	0.105	0.222	-0.122–0.330
KAN-Payne	$\log g$ (dex)	9117	0.197	0.199	0.000–0.418
TransformerPayne-style	$\log g$ (dex)	9117	0.173	0.193	-0.027–0.372
Payne-MLP	[Fe/H] (dex)	9117	0.048	0.088	-0.032–0.152
KAN-Payne	[Fe/H] (dex)	9117	0.037	0.053	-0.015–0.095
TransformerPayne-style	[Fe/H] (dex)	9117	0.035	0.061	-0.020–0.105

KAN-Payne shows the smallest robust scatter in $T_{\rm eff}$ and [Fe/H] and also keeps the largest model-specific atmospheric-quality sample before the intersection cut. On the common sample its [Fe/H] scatter is 0.053 dex, compared with 0.088 dex for Payne-MLP and 0.061 dex for the TransformerPayne-style baseline. The $T_{\rm eff}$ scatter is 113 K for KAN-Payne, 125 K for Payne-MLP, and 136 K for the TransformerPayne-style baseline. The main remaining systematic is a positive $\log g$ offset of 0.20 dex, slightly larger than the Payne-MLP offset and comparable to the TransformerPayne-style baseline. This offset is expected to be sensitive to continuum placement, the adopted giant-star selection, and the fact that all three fits optimize the full 25-label Payne vector although only three atmospheric labels are reported here.

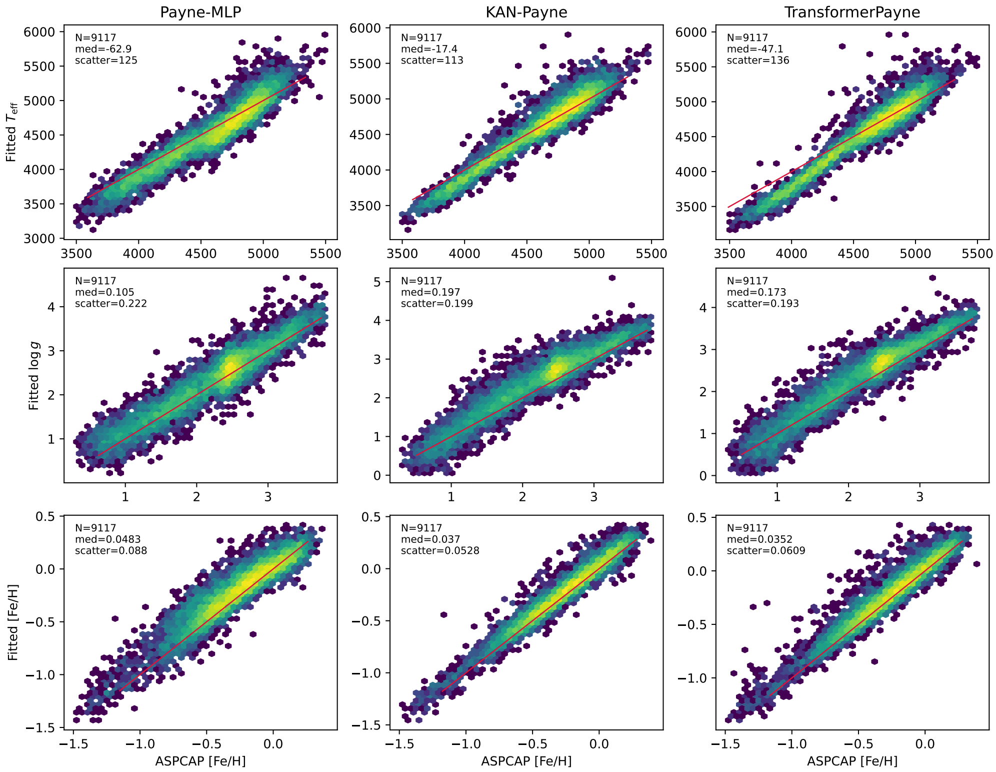

Figure 5. APOGEE DR17 atmospheric-label comparison with ASPCAP. Each column shows one emulator and each row shows one reported atmospheric label, using the same 9117-star common quality sample as Table 2. The red line marks the one-to-one relation.

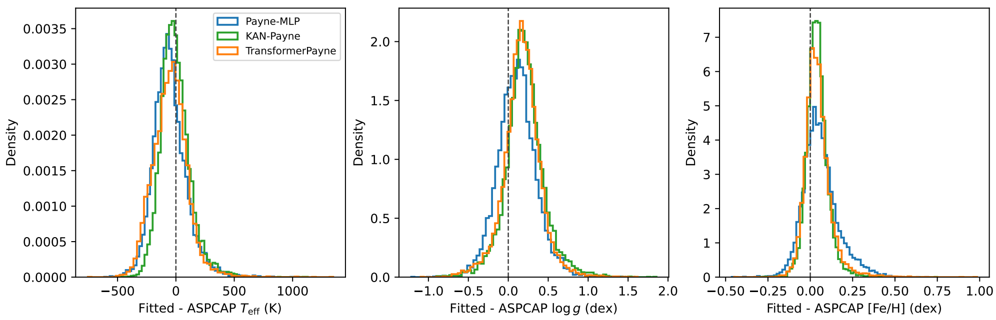

Figure 6. APOGEE DR17 residual distributions for the atmospheric labels. The narrower KAN-Payne [Fe/H] distribution is consistent with the synthetic-grid validation result and indicates that the KAN emulator improves the forward model in abundance-sensitive spectral regions on the common quality sample.

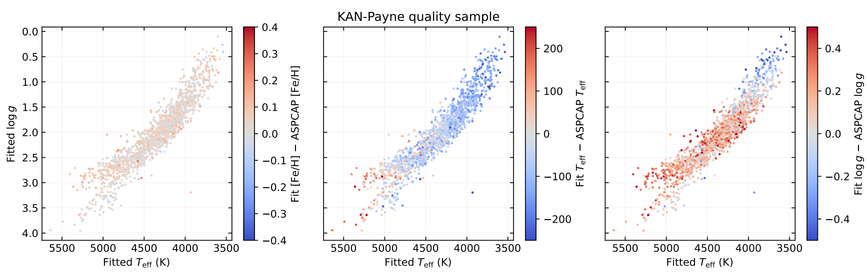

Figure 7. KAN-Payne APOGEE DR17 Kiel-diagram residual diagnostics. The panels show the clean red-giant sample in the ASPCAP $T_{\rm eff}$–$\log g$ plane and color the same stars by fitted-minus-ASPCAP residuals in [Fe/H], $T_{\rm eff}$, and $\log g$. The color limits are ± 0.4 dex, ± 250 K, and ± 0.5 dex, respectively.

5. Discussion

This benchmark separates emulator accuracy from downstream label inference. KAN-Payne is designed for the part of the problem where The Payne is already attractive: repeated differentiable full-spectrum evaluations in a moderate-dimensional label space. Its main competitor in that setting is the Payne-MLP. TransformerPayne-style emulators address a related but not identical trade-off: they model flux as a function of labels and wavelength and can exploit long-range correlations, but validation and fitting require scanning wavelength chunks. The comparison is therefore not a single ranking of neural architectures. It identifies which architecture is preferable for a specific survey task: full-spectrum fitting, wavelength-conditioned scalar emulation, or direct compatibility with existing Payne workflows.

The controlled synthetic benchmark is the cleanest evidence for the KAN-Payne architecture. All three trainable models see the same 800 training spectra, 200 validation spectra, label scaling, wavelength grid, and loss definition. Under these conditions KAN-Payne reduces the validation MAE by 63.8% relative to the controlled Payne-MLP and by 40.6% relative to the TransformerPayne-style baseline. The public pretrained Payne MLP still gives the lowest synthetic validation MAE on this exact file, so it is retained as an external reference rather than treated as a controlled baseline. The controlled comparison in this paper is therefore deliberately narrower: it asks how the local KAN, MLP, and TransformerPayne-style implementations behave when the data loader, split, label scaling, target spectra, and optimization setup are held fixed.

The APOGEE experiment is a more demanding test because the synthetic and observed spectra do not share the same continuum placement, line-spread function, noise model, and abundance system. We therefore treat agreement with ASPCAP as an external consistency check rather than an absolute accuracy measurement. The common-sample comparison shows that the improvement in synthetic emulation carries into the observed-spectrum fits most clearly for [Fe/H], where KAN-Payne gives the narrowest residual distribution. The $T_{\rm eff}$ scatter also improves relative to both baselines. The positive $\log g$ offset remains the main systematic residual and indicates that a production catalog would need a more careful treatment of gravity calibration, continuum placement, and possibly empirical post-calibration against clusters or asteroseismic gravities. For KAN-Payne on the common sample, the $\log g$ residual has only weak rank correlations with ASPCAP $T_{\rm eff}$, $\log g$, [Fe/H], and median S/N, with Spearman coefficients of 0.22, 0.06, -0.04, and -0.09, respectively. A linear fit against ASPCAP $\log g$ gives a slope of only 0.038 dex per dex, so the offset is closer to a broad zero-point displacement than to a steep trend across the giant branch.

The near equality of the APOGEE spectral MAE for Payne-MLP and KAN-Payne should not be read as a contradiction of the label comparison. A full-spectrum uncertainty-weighted objective can be dominated by continuum-level residuals, weak blends, and pixels whose label sensitivity is low. The atmospheric labels reported in Table 2, by contrast, are constrained by specific spectral regions and by the local curvature of the emulator with respect to the fitted labels. KAN-Payne can therefore give a similar total flux residual while producing a tighter [Fe/H] or $T_{\rm eff}$ residual distribution against ASPCAP.

The fitting experiment uses ASPCAP atmospheric labels to initialize the optimizer when they are available. This makes the comparison stable and prevents failed fits from dominating the first survey-scale test, but it also means the APOGEE run is not yet a blind inference experiment. As a direct check on this issue, we repeated the first 1000-star fits for the two vector-output emulators from the midpoint of the synthetic training grid rather than from ASPCAP. For KAN-Payne the median $\chi^2_{\rm red-like}$ changes only from 3.922 to 3.912 and the median spectral MAE changes from 142.31 to 142.54 in units of 10^4 times the flux residual. The corresponding Payne-MLP values change from 3.955 to 3.951

and from 143.59 to 144.28. The $\chi^2_{\mathrm{red-like}}$ and spectral MAE therefore change by less than 0.3% between ASPCAP initialization and midpoint initialization for both vector-output models. At the label level, KAN-Payne midpoint-init gives T_{eff}, $\log g$, and [Fe/H] scatters of 108.4 K, 0.222 dex, and 0.059 dex, compared with 117.9 K, 0.211 dex, and 0.056 dex for ASPCAP-init on the same 1000 stars. The direct midpoint-minus-ASPCAP-init scatter is only 14.6 K, 0.019 dex, and 0.011 dex for these three labels. The corresponding test for the TransformerPayne-style baseline is deferred to a follow-up technical note because its per-spectrum fit cost is higher under wavelength-chunked evaluation. These numbers do not make the APOGEE experiment fully blind, but they show that the vector-output comparison is not driven only by the ASPCAP starting point.

The continuum treatment is another limitation. The Payne APOGEE fitting code uses a fixed set of continuum pixels, whereas the present implementation reconstructs an equivalent continuum mask from the public synthetic grid by selecting stable pixels close to unity flux in each detector segment. The result is adequate for a method comparison, but it is not identical to the original Payne DR14 normalization. Future catalog production should carry the continuum mask as a versioned data product and should test whether a joint continuum–label fit reduces the residual gravity trends. The current 25-label fit also includes macroturbulence and $^{12}C/^{13}C$. At APOGEE resolution, macroturbulence is partly degenerate with rotational broadening and line-spread-function mismatch, while $^{12}C/^{13}C$ is weak for many stars in this giant sample. These two labels should therefore be treated as nuisance dimensions in the present method comparison rather than as validated catalog quantities.

Finally, the TransformerPayne-style entry in this paper is a matched local baseline, not an exhaustive optimization of transformer architectures. The same is true for the KAN configuration. The present result establishes that KAN layers are a competitive replacement for the Payne MLP under a fixed training protocol. It does not prove that the adopted hidden widths, spline settings, or optimizer schedule are globally optimal.

6. Conclusions

We introduce KAN-Payne as a KAN-based spectral emulator for Payne-style stellar label inference. The method keeps the differentiable full-spectrum structure of The Payne while replacing the fixed-activation MLP approximation with adaptive KAN layers. On the public Payne APOGEE-like synthetic grid, KAN-Payne reaches a validation MAE of 20.89 and RMSE of 57.12 in units of 10^4 times the flux residual, outperforming the controlled Payne-MLP and TransformerPayne-style baselines under the same split and loss. In the APOGEE DR17 10000-star experiment, the common 9117-star quality sample shows that KAN-Payne gives the smallest scatter in T_{eff} and [Fe/H] relative to ASPCAP, with respective robust scatters of 113 K and 0.053 dex.

These results support KAN-Payne as a useful vector-output alternative to the original Payne MLP and wavelength-conditioned transformer emulators. It is more accurate than the MLP in the controlled synthetic test and remains compatible with full-spectrum gradient-based fitting. At the same time, the APOGEE experiment shows that careful observed-spectrum normalization and initialization tests remain essential before releasing a production stellar parameter catalog.

Funding: This research was funded by the National Key R&D Program of China, grant number 2025YFF0511001, and by the National Astronomical Observatories, Chinese Academy of Sciences, under Project No. E4ZR050901.

Acknowledgments: This work made use of public data from the Sloan Digital Sky Survey IV APOGEE program and public synthetic spectra and code resources from The Payne and TransformerPayne projects. The authors thank the maintainers of these data products and software tools. The numerical experiments were carried out using local computing resources and the `lily` GPU server.

Data Availability Statement: The APOGEE DR17 spectra and ASPCAP catalog used in this work are publicly available from the SDSS Science Archive Server. The Payne synthetic spectra are available from the public Payne repository. The analysis code is available at https://github.com/wangruibistu/kan_payne. The data package prepared for this manuscript is available from Zenodo at https://doi.org/10.5281/zenodo.20782050. It contains the 576-pixel continuum mask, training configuration and history files, validation metrics, APOGEE fit summaries, trained checkpoints, and the small diagnostic tables used in the revision.

Conflicts of Interest: The authors declare no conflicts of interest.

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